CSE4088 Introduction to Machine Learning

KNN classifier

K-NN classifier

- Assign the input to the class having the most examples among the k neighbors of the input.
- All neighbors have equal vote, and the class having the maximum number of voters among the k neighbors is chosen.
- Special case is the nearest neighbor classifier where k = 1.
  - The input is assigned to the class of the nearest pattern.
  - This divided the space in the form of a Voronio tesellation.
  - Suboptimal procedure: lead to an error greater than the minimum possible, the Bayes rate. (with unlimited number of prototypes the error rate is never worse than twice the Bayes rate.

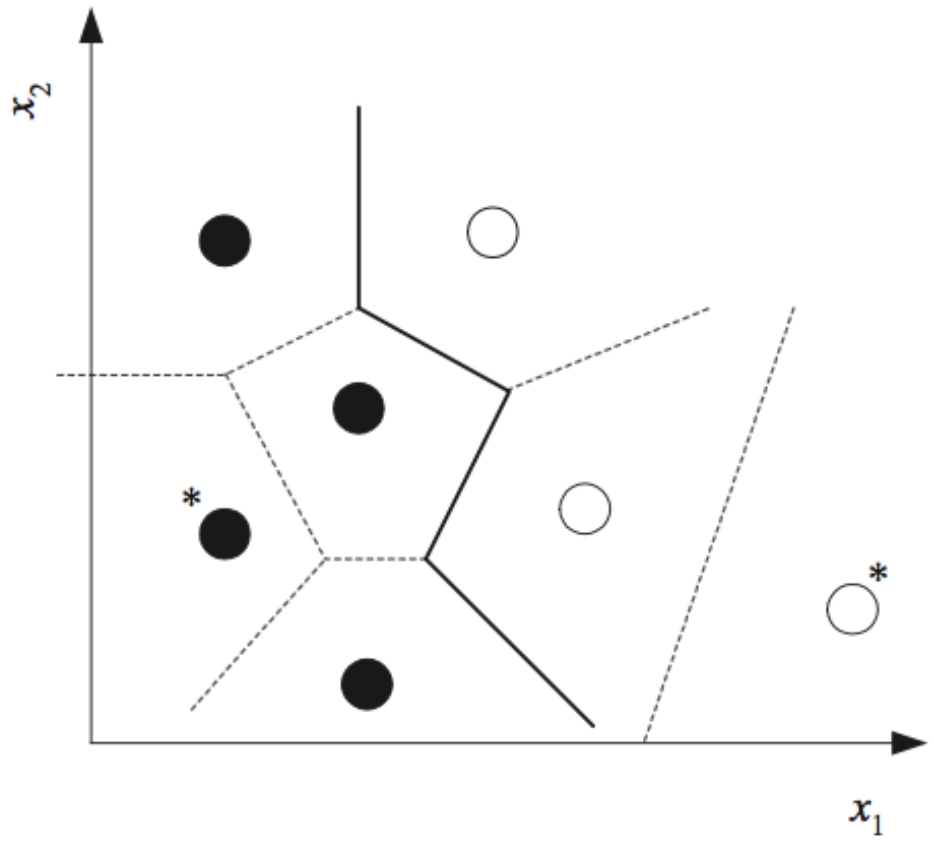
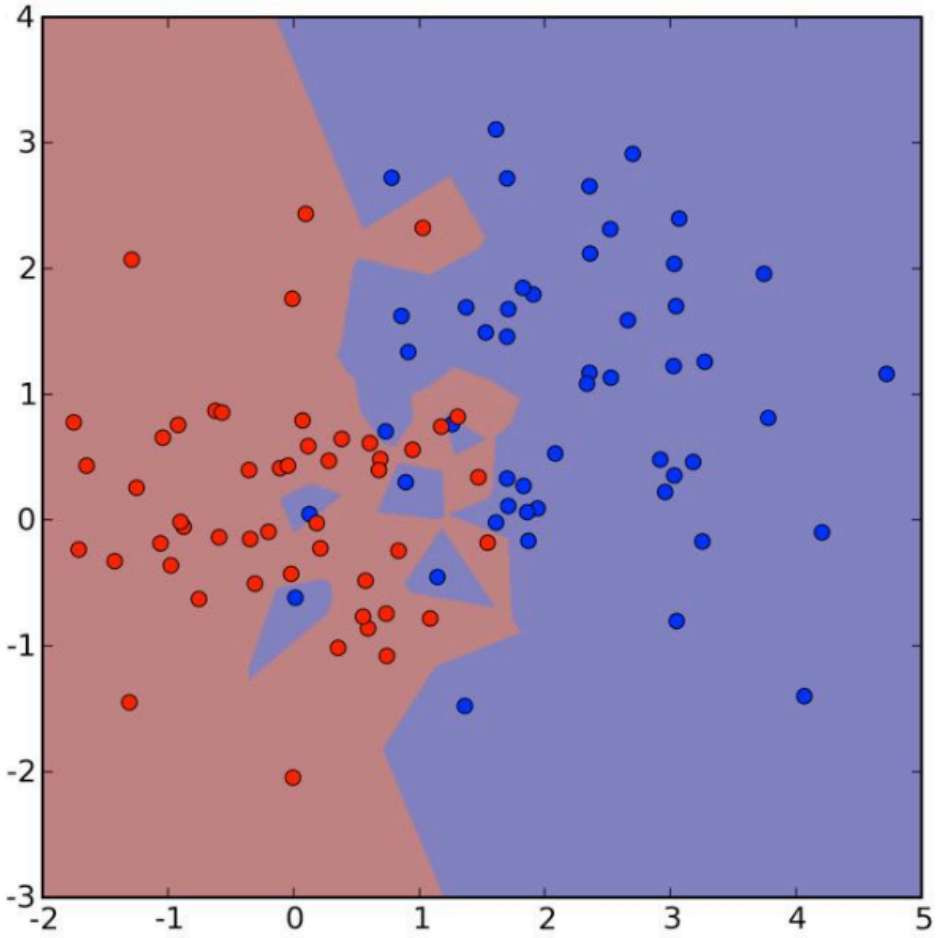


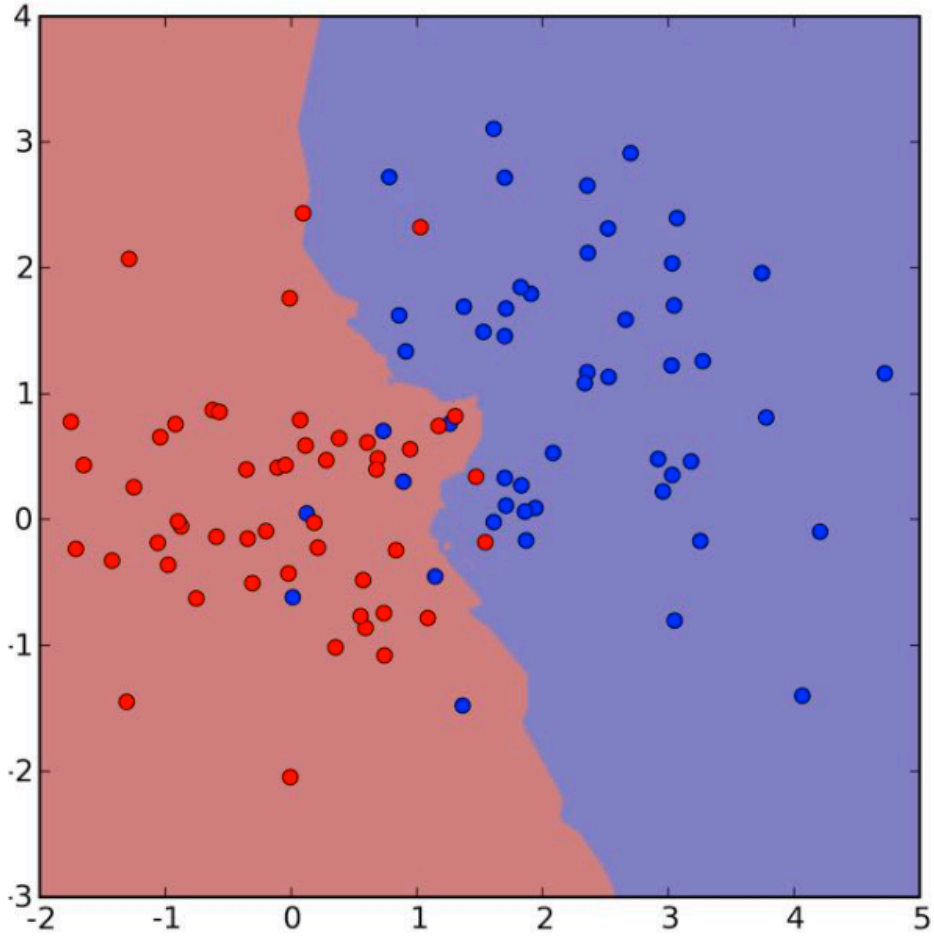
Figure 8.5 Dotted lines are the Voronoi tesselation and the straight line is the class discriminant. In condensed nearest neighbor, those instances that do not participate in defining the discriminant (marked by '*') can be removed without increasing the training error.

1-Nearest Neighbor



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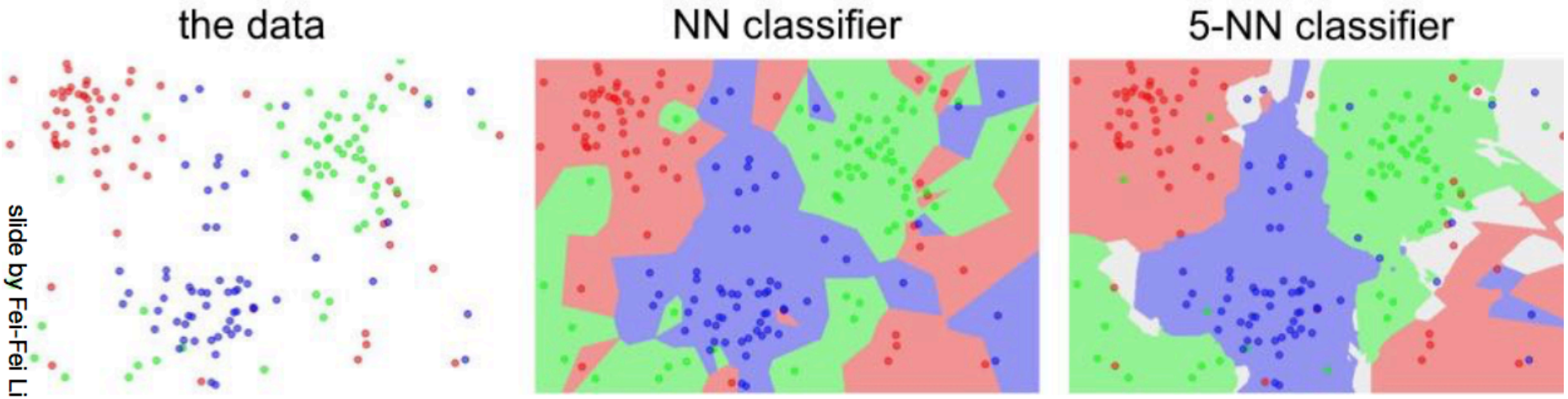
4-Nearest Neighbors Sign



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k-Nearest Neighbor

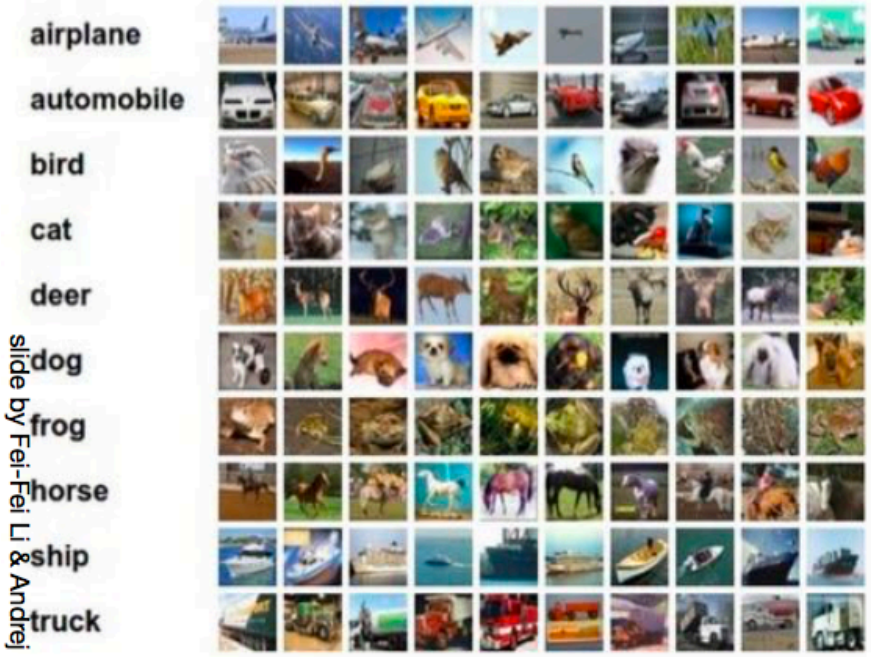
find the k nearest images, have them vote on the label



http://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

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Example dataset: **CIFAR-10**
10 labels
50,000 training images
10,000 test images.



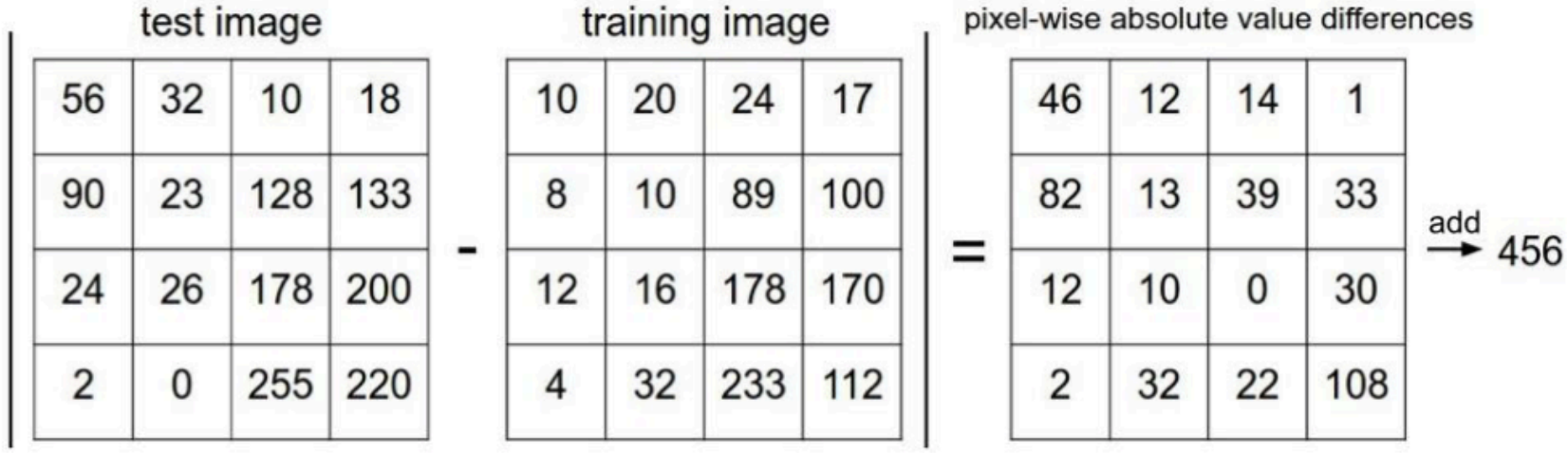
For every test image (first column), examples of nearest neighbors in rows



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How do we compare the images? What is the **distance metric**?

L1 distance: $d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$



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The choice of distance is a **hyperparameter**
common choices:

L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p \left(I_1^p - I_2^p\right)^2}$$

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Metrics and NN Classifier

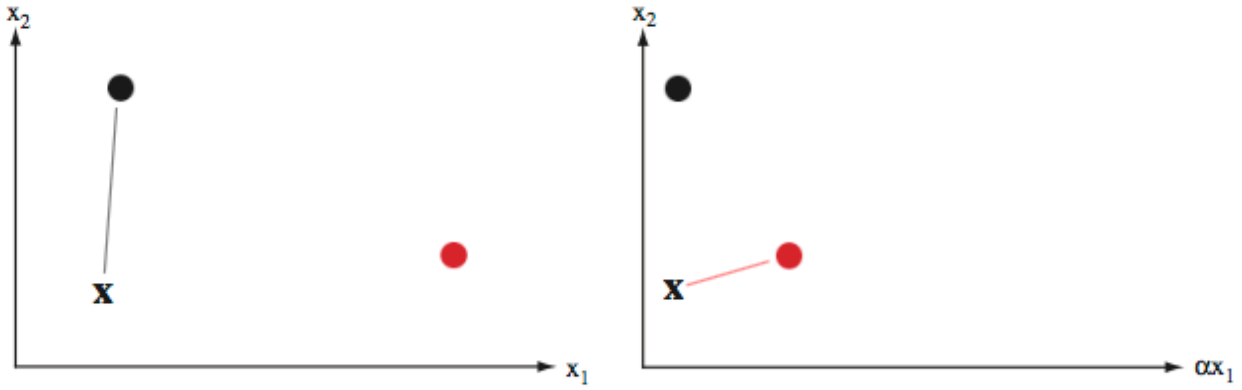


Figure 4.18: Even if each coordinate is scaled by some constant, the resulting space still obeys the properties of a metric. However, a nearest-neighbor classifier would have different results depending upon such rescaling. Consider the test point $\mathbf{x}$ and its nearest neighbor. In the original space (left), the black prototype is closest. In the figure at the right, the x_1 axis has been rescaled by a factor $1/3$; now the nearest prototype is the red one. If there is a large disparity in the ranges of the full data in each dimension, a common procedure is to rescale all the data to equalize such ranges, and this is equivalent to changing the metric in the original space.

Outlier Detection

- An **outlier**, **novelty** or **anomaly** is an instance that is very much different from other instances in the sample. Spot what normally does not happen.
- Outlier may indicate an abnormal behavior of the system:
  - Fraud credit card transaction
  - A tumor in an image
  - Network intrusion attempts
  - Recording errors due to faulty sensors
- Is not a two-class classification problem because there are very few outlier instances.
- It is the typical instances that are modeled (one-class classification).
  - Any instance that does not fit the model (e.g. having a low probability) is an anomaly.

Outlier Detection

- **Problem**: data used to train the outlier detector is unlabeled and may contain outliers.
  - Better to use a nonparametric density estimator
  - E.g. A single outlier may seriously corrupt the estimate mean and covariance of a Gaussian.

Local Outlier Factor

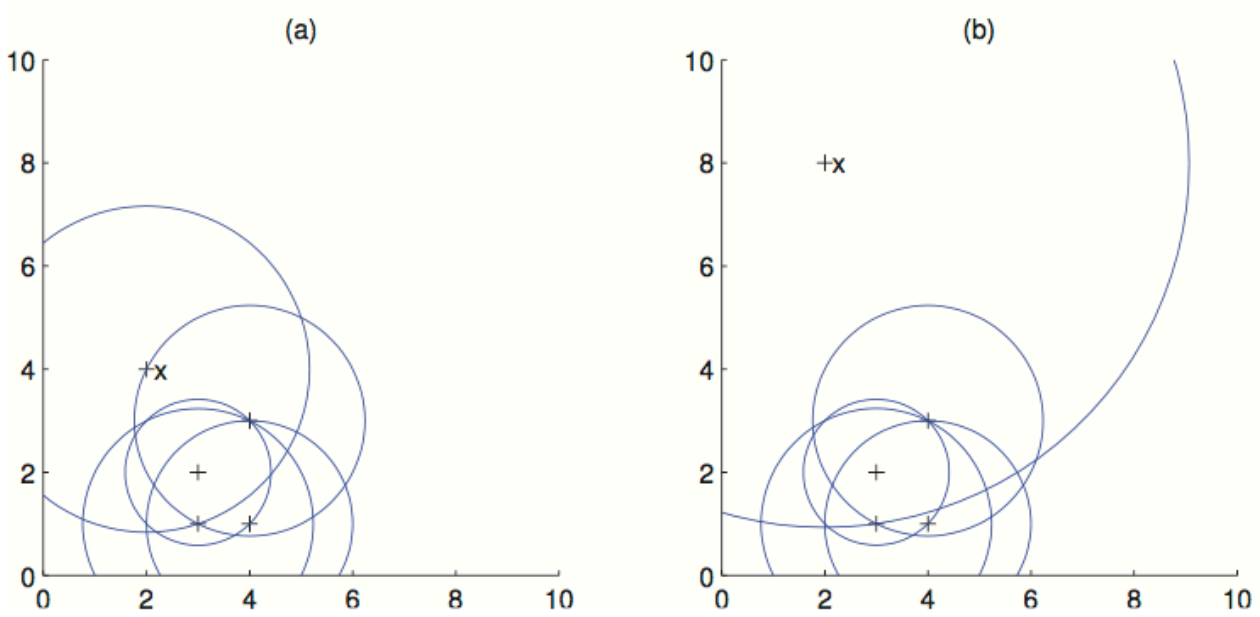


Figure 8.8 Training instances are shown by '+', '×' is the query, and the radius of the circle centered at an instance is equal to the distance to the third nearest neighbor. (a) LOF of '×' is close to 1 and it is not an outlier. (b) LOF of '×' is much larger than 1 and it is likely to be an outlier.

$$\mathrm{LOF}(\boldsymbol{x}) = \frac{d_k(\boldsymbol{x})}{\sum_{\boldsymbol{s}\in\mathcal{N}(\boldsymbol{x})} d_k(\boldsymbol{s}) / |\mathcal{N}(\boldsymbol{x})|}$$

$d_k(\boldsymbol{x})$: distance between instance x and its k-th nearest neighbor

$\mathcal{N}(\boldsymbol{x})$: set of k nearest neighbors of x.